

Vibration Excessive

Symptoms

- Vibration excessive
- Cab noise due to vibration

How To Use This Tree

This symptom tree can be used to troubleshoot all vibration-based symptoms listed above. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptoms. Perform the list of troubleshooting steps in the sequence shown in the Specifications/Repair section of the tree.

Shoptalk

Vibration Troubleshooting Documentation Information Questions

- 1) What is the original equipment manufacturer (OEM) make/model?
- 2) What are the mileage/hours?
- 3) Has there been any recent repair and/or maintenance history?
 - Any engine/clutch/transmission rebuild, removal, and installation?
 - History of repeatedly broken brackets and/or capscrews (alternator, fan, exhaust, etc.)?
 - Structural modifications to the vehicle from OEM built?
- 4) Description of vibration?
 - What is shaking (mirror, seat, steering wheel, cab/dash, etc.)?
 - Is there excessive noise in the cab during the vibration?
- 5) What are the conditions when the complaint occurs?
 - Power take-off (PTO)?
 - Power output (hard pull, during lug down, etc.)?
 - During acceleration and/or deceleration?
 - At idle?
 - With or without a trailer?
 - Does vibration increase with engine speed?
 - Does vibration increase with road speed?
 - Is the vibration at a certain engine revolutions per minute (rpm)?

6) Has the vibration been present since new? (From new, recent repair, modification to equipment)

7) Can the vibration be easily duplicated?

8) Do you have another piece of equipment with the same specification which exhibits the same complaint? (If yes, get the engine serial number (ESN) and possibly test the vehicle)

9) Are you the **only** operator of the equipment? (If no, are the symptoms noticed by other operators)?

General Information

Vibration complaints can be very difficult to troubleshoot and understand the root cause. This troubleshooting document was designed to help guide you through the logical steps of identifying the source. Vibration acceptance is very subjective; what is objectionable to one person can possibly be acceptable to another.

Vibration complaints can be caused by many parts in the system (system includes the engine, driven component, mounts, and equipment). The cause can be transmitted or generated from a remote point that is **not** readily apparent.

Cummins Inc. experience has shown that the engine is rarely the cause of an operator complaint. The majority of the time, it is the engine mounts or design of the various components on the equipment. The engine is **only** at fault if there is a misfire or an engine component that is out of balance.

Vibration complaints that occur **only** at idle speed are most likely caused by the engine mounts. If the engine mount natural frequency is close to the engine firing frequency, the engine will cause the mounts to amplify the normal vibration on an engine idling and cause the adjacent components to vibrate excessively.

Natural Frequency

- Natural frequency, as the name implies, is the frequency at which an object wants to naturally vibrate. The frequency is primarily dependent on mass and elasticity.

Types of Vibration

1) Linear

a) Rotating components

b) Torque reaction

- Caused by unbalanced rotating components and cylinder firing impulses.
- Can be felt and observed visibly.
- When excessive, can cause operator discomfort and destruction of components.

2) Torsional - twisting stresses

- Cyclic speeding and slowing of rotating components.
- Controlled by flywheel mass and vibration damper.

- Can **NOT** be felt by the operator.
- Can damage gears and splines.

3) Resonant - component excited at natural frequency

- Is actually linear vibration.
- Resonant vibration occurs when a system or component is excited by linear vibration at its natural frequency.
- Vibration will increase in amplitude as the system's natural frequency is approached. Amplitude will decrease as the exciting forces (engine firing frequency) increase in frequency beyond the system's natural frequency.
- Resonant vibration can be many times larger in amplitude than the exciting force.
- Vibration **must** be controlled by design of mounts (engine and cab) and components.

Engine and Cab Mounts

- The mounts **must** be designed to isolate or reduce the transmission of engine and equipment component vibrations.
- For maximum isolation, it is desired that the natural frequency of the mount be as low as possible.
- Good engine mounts will reduce the amount of engine vibration transmitted to the chassis frame by at least 50 percent at idle.
- Hard engine mounts will give little or no isolation, and can actually magnify the vibration transmitted to the chassis.
- Stiffness (durometer) and size of the isolator, along with the weight of the engine or component applied, are the determining factors when designing a mounting system. An isolator that is correct for one engine, can possibly **not** be right for another. Likewise, because of weight differential, a particular isolator designed for the rear of an engine, probably will **not** be ideal for the front.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Perform the basic troubleshooting procedures.	
	STEP 1A. Document the information questions in the Shop Talk section of this tree.	Documentation completed?
	STEP 1B. Duplicate the complaint based on the customer description.	Customer's complaint be duplicated?

STEPS		SPECIFICATIONS
	STEP 1C. Check for active fault codes or high counts of inactive fault codes.	Active fault codes or high counts of inactive fault codes?
	STEP 1D. Determine if the engine is running rough.	Engine running rough?
	STEP 1E. Marine application.	Engine installed in a marine application?
STEP 2.	Inspect engine brake adjustment.	
	STEP 2A. Verify the engine brake lash measurement (if equipped).	Engine brake lash measurement correct?
STEP 3.	Inspect engine mounts.	
	STEP 3A. Perform a visual inspection of the engine mounts (without removal).	Visible engine mount damage?
	STEP 3B. Check for an engine mounted component contacting the frame or body.	Engine mounted components touching the frame or body?
	STEP 3C. Inspect the engine mounts.	All steps verified to be correct?
STEP 4.	Perform drive belt checks.	
	STEP 4A. Inspect the front end belt driven accessories.	Front end belt driven accessories damaged?
STEP 5.	Check vibration damper.	
	STEP 5A. Check for a damaged vibration damper.	Vibration damper damaged or out of specification?
STEP 6.	Marine applications.	
	STEP 6A. Check the gear ratio and propeller configuration.	Gear ratio and the propeller incorrectly matched to the engine power?
	STEP 6B. Check for the correct engine mounting isolators and for proper installation requirements.	Engine mount isolators correct and installed correctly?
	STEP 6C. Check for damaged engine mounts and isolators.	Engine mounts and isolators in good condition?
	STEP 6D. Check the engine driven accessories.	Engine driven accessory malfunctioning?

STEPS		SPECIFICATIONS
	STEP 6E. Check the shaft coupling to gear coupling alignment.	Shaft coupling to gear coupling misaligned?
	STEP 6F. Check the propeller shaft for proper installation.	Propeller shaft installed correctly?
	STEP 6G. Check the propeller shaft for straightness.	Propeller shaft straightness within the OEM specification?

STEP 1. Perform the basic troubleshooting procedures.

STEP 1A. Document the information questions in the Shop Talk section of this tree.

Conditions : None		
Action	Specification/Repair	Next Step
Perform the basic troubleshooting questionnaire. Complete the vibration troubleshooting documentation information questions contained in the Shop Talk section of this procedure.	Documentation completed? YES	1B
	Documentation completed? NO Repair : Complete the documentation.	1A

STEP 1B. Duplicate the complaint based on the customer description.

Conditions : None.		
Action	Specification/Repair	Next Step
Operate the equipment based on the description from the customer to duplicate the complaint. N/A	Customer's complaint be duplicated? YES	1C
	Customer's complaint be duplicated? NO	Repair complete.

STEP 1C. Check for active fault codes or high counts of inactive fault codes.**Conditions :**

- Turn keyswitch ON.
- Connect the recommended Cummins® electronic service tool, or equivalent.

Action	Specification/Repair	Next Step
Check the fault codes. • Use the electronic service tool to read the fault codes.	Active fault codes or high counts of inactive fault codes? YES Repair : See the corresponding Electronic Control System Troubleshooting and Repair manual for the engine being serviced.	Repair complete.
	Active fault codes or high counts of inactive fault codes? NO	1D

STEP 1D. Determine if the engine is running rough.**Conditions :**

- Operate engine at idle speed (less than 900 rpm).
- Turn accessories OFF(air conditioning, fan, PTO).
- Operate engine at operating temperature (greater than 170°C [70°F]).

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Determine if the engine is running rough at engine idle. Refer to the Engine Runs Rough troubleshooting symptom tree in Section TS or the Engine Performance Troubleshooting Tree in Section TT.	Engine running rough? YES Repair : Refer to the Engine Runs Rough troubleshooting symptom tree in Section TS or the Engine Performance Troubleshooting Tree in Section TT.	Complete Engine Runs Rough troubleshooting tree
	Engine running rough? NO	1E

STEP 1E. Marine application.

Conditions : • None.		
Action	Specification/Repair	Next Step
Engine in a marine application? N/A	Engine installed in a marine application? YES	5A
	Engine installed in a marine application? NO	2A

STEP 2. Inspect engine brake adjustment.

STEP 2A. Verify the engine brake lash measurement.

Conditions : • None.		
Action	Specification/Repair	Next Step

Verify the engine brake lash measurement, if equipped. Reference Procedure 020-004 in Section 20 of the appropriate service manual. N/A	Engine brake lash measurement correct? YES	3A
	Engine brake lash measurement correct? NO Repair : Adjust the engine brake lash measurement to the correct lash setting. Reference Procedure 020-004 in Section 20 of the appropriate service manual.	Repair complete.

STEP 3. Inspect engine mounts.

STEP 3A. Perform a visual inspection of the engine mounts (without removal).

Conditions :

- Do **not** operate engine.
- Install engine mounts.

Action	Specification/Repair	Next Step
Perform a visual inspection of the engine mounts. Look for obvious damage or something shorting against the mounts, preventing isolation. A more detail inspection will be carried out later in the procedure.	Visible engine mount damage? YES Repair : Repair or replace the engine mounts. Reference Procedure 016-010 in Section 16 of the appropriate service manual.	Repair complete.
	Visible engine mount damage? NO	3B

STEP 3B. Check for an engine mounted component contacting the frame or body.

Conditions :

- Do **not** operate engine.
- Install engine mounts.

Action	Specification/Repair	Next Step
Check for an engine mounted component touching the frame or body. Inspect the engine and engine mounted components to make sure none of them are touching the frame and/or body. Including but not limited to the following: • Clamps • Mounting hardware • Exhaust system • Air intake piping • Cooling package support.	Engine mounted components touching the frame or body? YES Repair : Correct the mounting of the engine mounted component.	Repair complete.
	Engine mounted components touching the frame or body? NO	3C

STEP 3C. Inspect the engine mounts.

Conditions :

- Do **not** operate engine.
- Install engine mounts.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Perform a detailed inspection of the engine mount brackets, isolators, and mounting hardware. • Check the engine mount isolators for installation damage. • Check the alignment of the engine mount brackets. • Check for premature wear on the engine mount isolators and mounting hardware.	All steps verified to be correct? YES	4A
	All steps verified to be correct? NO Repair : Repair or replace the damaged components.	Repair complete.

STEP 4. Perform drive belt checks.

STEP 4A. Check front end belt driven accessories.

Conditions : • Turn keyswitch OFF.		
Action	Specification/Repair	Next Step

Inspect the following components for damage that can cause engine noise or vibration. Reference the procedures in the appropriate service manual or the equipment manufacturer service information. • Charging system alternator - Reference Procedure 013-001 in Section 13 of the appropriate service manual or the equipment manufacturer service information. • Automatic charging system alternator belt tensioner, if equipped - Reference Procedure 013-021 in Section 13 of the appropriate service manual. • Cooling fan hub assembly - Reference Procedure 008-036 in Section 8 of the appropriate service manual or the equipment manufacturer service information. • Cooling fan and cooling fan blades - Reference Procedure 008-040 in Section 8 of the appropriate service manual or the equipment manufacturer service information.	Damaged component? YES Repair : Repair or replace the malfunctioning cooling belt drive accessory.	Repair complete.
--	--	--------------------------------

- | | | |
|---|--|--|
| <ul style="list-style-type: none">• Automatic water pump belt tensioner, if equipped - Reference Procedure 008-080 in Section 8 of the appropriate service manual.• Automatic cooling fan belt tensioner, if equipped - Reference Procedure 008-087 in Section 8 of the appropriate service manual.• Cooling fan belt - Reference Procedure 008-002 in Section 8 of the appropriate service manual or the equipment manufacturer service information.• Water pump - Reference Procedure 008-062 in Section 8 of the appropriate service manual.• Pulley alignment from fan pulley and crankshaft pulley to other front-end mounted grove pulleys, if equipped - Reference Procedure 001-022 in Section 1 of the appropriate service manual.• A/C compressor and water pump belt alignment, if equipped - Reference Procedure 009-052 in Section 9 of | | |
|---|--|--|

the appropriate service manual.

- Idler pulley, if equipped. Reference Procedure 009-054 in Section 9 of the appropriate service manual.
- Check alignment of the Alternator/Refrigerant Compressor Mounting Bracket, if equipped. Reference Procedure 009-056 in Section 9 of the appropriate service manual.

Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19.

(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)

Damaged component?
NO

5A

STEP 5. Check vibration damper.

STEP 5A. Check for a damaged vibration damper.

Conditions :

- Do **not** operate engine.

Action

Specification/Repair

Next Step

Remove and visually inspect the vibration damper. Reference Procedure 001-052 in Section 1 of the appropriate service manual for vibration damper inspection specifications.	Vibration damper damaged or out of specification? YES	Repair complete.
	Vibration damper damaged or out of specification? NO	Contact a Cummins® technical support specialist or the equipment manufacturer.

STEP 6. Marine applications.

STEP 6A. Check the gear ratio and propeller configuration.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for an incorrect matching of the gear ratio and propeller to the engine power. N/A	Gear ratio and the propeller incorrectly matched to the engine power? YES Repair : Contact a Cummins® Distributor or a Marine District Field Service Manager.	Repair complete.
	Gear ratio and the propeller incorrectly matched to the engine power? NO	6B

STEP 6B. Check for the correct engine mounting isolators and for proper installation requirements.

Conditions :

- None.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check for the correct engine mount isolators and for propeller installation requirements. N/A	Engine mount isolators correct and installed correctly? YES	6C
	Engine mount isolators correct and installed correctly? NO Repair : Check for proper isolator installation requirements. Replace and repair vibration isolators as needed. Reference Procedure 016-026 in Section 16 of the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Recreational Installation Directions, Bulletin 3884649. If the isolators are not manufactured by Cummins Inc., see equipment manufacturer service information.	Repair complete.

STEP 6C. Check for damaged engine mounts and isolators.

Conditions : • None.		
Action	Specification/Repair	Next Step

Inspect the engine mount and isolators for damage. N/A	Engine mounts and isolators in good condition? YES	6D
	Engine mounts and isolators in good condition? NO Repair : Remove and replace the engine mount isolators. Reference Procedure 016-026 in Section 16 of the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Recreational Installation Directions, Bulletin 3884649. If the isolators are not manufactured by Cummins Inc., see equipment manufacturer service information.	Repair complete.

STEP 6D. Check the engine driven accessories.

Conditions :

- Turn keyswitch ON.
- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check for engine driven accessory malfunctions. - Isolate or disconnect the accessories and check for vibration. - Do not operate the engine if the sea water pump is disconnected.	Engine driven accessory malfunctioning? YES Repair : Determine the cause of the malfunctioning accessories and correct the problem. See the Exhaust System section in the Marine Recreational Installation Directions, Bulletin 3884649, and the equipment manufacturer service information.	Repair complete.
	Engine driven accessory malfunctioning? NO	6E

STEP 6E. Check the shaft coupling to gear coupling alignment.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check the shaft coupling to gear coupling alignment. N/A	Shaft coupling to gear coupling misaligned? YES Repair : Repair or replace as needed. Reference Procedure 016-025 in Section 16 of the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Recreational Installation Directions, Bulletin 3884649, and gear manufacturer recommendations.	Repair complete.
	Shaft coupling to gear coupling misaligned? NO	6F

STEP 6F. Check the propeller shaft for proper installation.

Conditions : • None.		
Action	Specification/Repair	Next Step

Check the propeller shaft for proper installation. N/A	Propeller shaft installed correctly? YES	6G
	Propeller shaft installed correctly? NO Repair : Repair or replace as needed. Reference Procedure 016-025 in Section 16 of the appropriate service manual and the Engine Mounting/Drive Systems section in the Marine Recreational Installation Directions, Bulletin 3884649, and gear manufacturer recommendations.	Repair complete.

STEP 6G. Check the propeller shaft for straightness.

Conditions : • None.		
Action	Specification/Repair	Next Step
Check the propeller shaft for straightness. N/A	Propeller shaft straightness within the OEM specification? YES	Contact a Cummins® technical support specialist or the equipment manufacturer.
	Propeller shaft straightness within the OEM specification? NO Repair : Repair or replace the propeller shaft as needed. See equipment manufacturer service information.	Repair complete.

